

UnionLabs SDR Platform — Beginner’s Guide

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UnionLabs SDR Platform — Beginner’s Guide

A single, self-contained guide: what the platform is, how to run it, how to add your own algorithm, and how to drive the radio directly. No prior knowledge of the codebase needed.

PHY = the “modem”. It turns your data (bytes / a float32 vector) into radio waves and back: framing -> error-correction (FEC) -> modulation -> radio, and the reverse on receive. You bring an *algorithm*; the PHY moves its data over the air.

The promise. You write the algorithm once. The same file runs with no radio, over a USRP software-defined radio, or over a LoRa module — you change a flag, not your code.

1. Quick start

Install first — Python 3.8 or newer, one command, no radio and no C++ build:

```
pip install -r requirements.txt
./run.sh selftest           # confirm the install works (every experiment x every radio-free PHY)
```

If `selftest` prints all N checks passed, everything below will work. If only the `usrp` checks fail, the compiled `pyphy` extension does not match your Python — build it with `drivers/usrp/bindings/build.sh`, or just use `--channel ideal` / `--channel lora`, which need nothing.

```
./run.sh           # run an algorithm over the PHY (defaults: echo, no radio)
./run.sh list      # list the algorithms available
./run.sh --algo marl # run a specific algorithm
./run.sh --help    # help + every option
./radio.sh rx      # raw receive on a USRP (B210 default; --device n210|x310)
./radio.sh tx      # raw transmit
```

Anything you don't specify takes a **default** (algo=echo role=loopback channel=ideal steps=5). run.sh prints the exact command it runs (the >> line) so you learn the underlying call.

2. Where things live

- **experiments/** — **everything you run**, one folder per experiment, each with an app.py. Your own code and the worked examples (FL, decentralized learning, MARL, CLIP semantic comm, STC-AirComp, jammer) all live here — one place to look.
- **union/** — the middleware you code against: run_algo.py + phy_link.py (the uniform API) and driver.py (the contract every PHY implements).
- **drivers/** — the physical layers, one folder each:
 - usrp/build/sdr_system — the C++ radio engine (the modem); radio.sh runs raw TX/RX.
 - usrp/python/ — sdr.py / run.py / configs/ (direct radio), channel_sense.py (RF utils).
 - usrp/bindings/pyphy — the DSP blocks as numpy functions.
 - lora/ — the SX1276 LoRa PHY (firmware + the Python driver).
- **docs/** — every guide, reference and PDF, including this one.

Three ways to use the PHY, easiest first:

1. **Run an algorithm** with run.sh (§3).
2. **Add your own algorithm** (§4).
3. **Drive the radio directly** with sdr.py / radio.sh (§5), or compose the DSP blocks (§6).

3. Running an algorithm over the PHY

Your algorithm only says **what to transmit** and **what to receive** — no radio code. Drop it in experiments/<name>/app.py (copy experiments/_template/), then run it by name.

3.1 The three independent choices

Every run answers three separate questions. Keeping them apart is the single most useful thing to understand about this platform:

```
./run.sh  --algo fl          --channel lora          --role loopback
          └─ WHAT ──┐      └─ WHICH PHY ─┐        └─ WHICH PART am I ─┐
```

1. **--algo** — what you are running. Any folder in experiments/ (./run.sh list).
2. **--channel** — which physical layer carries the bytes. Your algorithm never knows.
3. **--role** — whether this process is the *whole network* or *one node of it*.

A fourth, --<phy>-backend, says how the PHY is attached. **Every default backend needs no hardware**, so you develop the whole experiment on a laptop and change one flag to go on air.

3.2 Choosing a PHY (--channel)

--channel	What it is	Needs hardware?
ideal (default)	lossless, in-process	no
usrp	the real C++ modem (OFDM/single-carrier, LDPC/turbo FEC, sync, CFO)	no, with --usrp-backend pyphy

--channel	What it is	Needs hardware?
lora	the SX1276 LoRa PHY (255-byte MTU, fragmentation + ARQ)	no, with --lora-backend sim

```
./run.sh --algo fl # ideal – is my logic right?
./run.sh --algo fl --channel usrp --snr-db 6 # how noisy can the link get?
./run.sh --algo fl --channel lora --lora-sf 9 # what does this cost on LoRa?
```

Older spellings still work: --channel sim = ideal, --channel pyphy = usrp.

3.3 Choosing a role (--role)

Group roles build the whole network in one process — this is how you develop:

--role	Shape	Extra flags
loopback (default)	two ends, one round-trip each step	—
chain	initiator -> relay(s) -> responder, every hop over the PHY	--relays 1
gossip	N peers exchanging over a graph, no server	--agents 6 --topology
multi	N agents contending for one access point (collisions are real)	--agents 4
aircomp	N sensors transmit at once; the air sums them	--agents 8

Node roles run ONE node in this process — this is how you deploy, one terminal or one computer per node:

--role	This process is	Extra flags
tx	the initiator (transmits, then listens)	--radio, peer host
rx	the responder (listens, then answers)	--radio
relay	a middle node: receives from upstream, re-transmits downstream	--down-host, --down-port
peer	one node of a decentralized network — both TX and RX, at different steps	--node K --agents N

```
# develop: the whole 6-peer network in one process
./run.sh --algo dl --role gossip --agents 6 --topology ring

# deploy: one terminal per node (--node K implies --role peer)
./run.sh --algo dl --node 0 --agents 3
./run.sh --algo dl --node 1 --agents 3
./run.sh --algo dl --node 2 --agents 3
```

An algorithm can name its own roles. fl calls them client/server, dl calls them initiator/responder/peer. ./run.sh list prints each algorithm's roles, and you type the algorithm's own name: ./run.sh --algo fl --role server. tx/rx always work too.

3.4 Every run.sh option

Core

Option	Default	What it does
--algo <name>	echo	which folder under experiments/ to run
--channel ideal usrp lora	ideal	which PHY carries the payloads
--role <role>	loopback	see §3.3; or any role the algorithm declares
--steps <n>	5	how many rounds/round-trips to run
--agents <n>	4	number of nodes for gossip / multi / aircomp

Option	Default	What it does
<code>--node <k></code>	—	run ONE node of a decentralized network (implies <code>--role peer</code>)
<code>--relays <n></code>	1	relay nodes in the middle of a chain
<code>--topology <t></code>	ring	gossip graph: ring, full, or an edge list 0-1, 1-2, 2-0
<code>--snr-db <dB></code>	8	link signal-to-noise ratio (both PHYs)

Shared radio options (any PHY)

Option	Default	What it does
<code>--freq <MHz></code>	915	centre frequency. Outside 902–928 MHz you get a US-band warning
<code>--arq <scheme></code>	stop-and-wait	retransmission policy (only one implemented so far)
<code>--max-attempts <n></code>	per PHY	give up on a chunk after this many un-ACKed sends (USRP 50, LoRa 8)

USRP options (`--channel usrp`) — we assemble this PHY, so every part is yours to choose

Option	Default	What it does
<code>--usrp-backend pyphy radio</code>	pyphy	in-process modem, or real radios (needs two hosts)
<code>--modulation <NAME></code>	QPSK	BPSK, QPSK, 8-PSK, 16-QAM, DBPSK, DQPSK (<code>--scheme</code> is the same flag)
<code>--fec conv ldpc turbo ""</code>	turbo	error-correcting code; "" turns coding off
<code>--samp-rate / --symbol-rate</code>	2e6 / 1e6	sample and symbol rate in Hz
<code>--tx-gain / --rx-gain</code>	70 / 30	transmit and receive gain in dB
<code>--ack-transport tcp rf</code>	tcp	where the ACK travels: a socket, or a second RF path
<code>--ack-timeout <ms></code>	3000	how long the sender waits for an ACK
<code>--radio serial=... addr=...</code>	—	which USRP this process owns (B210 by serial, N210/X310 by address)
<code>--tx-args / --rx-args</code>	""	as above, when one node has two radios

LoRa options (`--channel lora`) — the chip embeds modulation and CRC, so these are its knobs

Option	Default	What it does
<code>--lora-backend sim serial spi</code>	sim	no hardware / Arduino on USB / Pi with the radio on SPI
<code>--lora-sf <7..12></code>	9	spreading factor: higher reaches further, costs exponentially more airtime
<code>--lora-cr <5..8></code>	5	coding rate denominator (4/5 ... 4/8)
<code>--lora-bw 125000 250000 500000</code>	125000	bandwidth in Hz; doubling it halves airtime and costs ~3 dB sensitivity
<code>--lora-power <dBm></code>	14	transmit power
<code>--lora-port <dev></code>	—	serial backend: e.g. <code>/dev/ttyUSB0</code>
<code>--lora-verbose</code>	off	print fragments, retransmissions and airtime for every message

Multi-process options (node roles)

Option	Default	What it does
<code>--peers <h1,h2,...></code>	all localhost	one host per node id, when nodes are on different computers
<code>--peer-port <n></code>	5800	base TCP port; node k listens on <code>peer-port + k</code>
<code>--peer-link tcp wireless lora</code>	follows <code>--channel</code>	how decentralized peers exchange
<code>--ack-host / --net-host</code>	127.0.0.1	the peer host's IP (USRP tx/rx roles)

Passing a knob for a PHY you did not select prints a NOTE rather than being silently ignored — `--lora-sf` with `--channel usrp` tells you so.

3.5 Reading the output

```
[run_algo] loaded experiments/fl via make(role) binding
[run_algo] role 'client' -> PHY end 'tx'
    step 0: req_ber=0.0000 rep_ber=0.0000 delivered=True
[run_algo] loopback done: 12/12 round-trips delivered over channel=ideal
[run_algo] lora PHY: {'msgs': 4, 'frags': 416, 'retx': 0, 'lost': 0,
    'airtime_s': 165.09, 'sf': 7, 'arq': 'stop-and-wait'}
```

- **delivered** — the round-trip completed and the CRC was clean.
- **req_ber / rep_ber** — bit error rate each way (the USRP modem reports real errors; LoRa reports 0 because the chip only surfaces CRC-valid packets, so a bad frame is a *loss*).
- **The PHY line** — what the run actually cost. For LoRa this is the headline: frags is how many 247-byte pieces your payload needed, retx the retransmissions, and airtime_s the real Semtech airtime. A 25 kB model over two rounds costs **165 s at SF7 and 3727 s at SF12** — the same experiment, 22× the time on air.

To write your own app.py, see §4.

4. Add your own algorithm

This has a guide of its own, so it stays in one place rather than drifting between two:

-> [HOW_TO_ADD_ALGORITHM.md](#) (repo root, also as [docs/HOW_TO_ADD_ALGORITHM.pdf](#))

It covers, in order:

1. **Where to put it** — experiments/<name>/app.py, and --algo <name> matches the folder.
2. **One file or many** — many: app.py is only the bridge, and your own modules, sub-packages and data sit beside it.
3. **What app.py must provide** — make(role) returning an object with transmit() / receive(msg), plus optional spec and on_result(ack).
4. **Two ways to write it** — inline (copy experiments/_template/), or a ~10-line binding onto your existing, untouched code (copy experiments/plain_echo/).
5. **Roles** — the four node types (tx, rx, relay, peer), naming your own with ROLES, and learning which node you are with make(role, index, total).
6. **Running it** — the same file over every PHY, and as a multi-node network.

The shortest possible version:

```
# experiments/my_algo/app.py
import numpy as np

class MyAlgo:
    spec = ("float32", (8,))
    def __init__(self, role): self.role = role
    def transmit(self):      return np.ones(8, np.float32)  # what to transmit (None = done)
    def receive(self, msg):  print("got", msg)              # what to receive

def make(role): return MyAlgo(role)

./run.sh --algo my_algo
```

5. Driving the radio directly (sdr.py / radio.sh)

When you want raw control of the modem (no algorithm layer), use the roles.

Shortcut: `./radio.sh tx` or `./radio.sh rx` runs a raw TX/RX on a USRP with the right per-device defaults — **B210 is the default**; add `--device n210` or `--device x310`. E.g. `./radio.sh rx --device n210 --freq 915e6`. Add `--dry-run` to just print the command.

From Python (`drivers/usrp/python/sdr.py`)

```
from sdr import tx, rx, both, source_arq, sink_arq, run_pair, options
options() # print every option + default
tx(tx_args="serial=30CD424", scheme="QPSK", tx_gain=78, fec=True).run() # transmit only
rx(rx_args="serial=30CD3F7", scheme="QPSK", fec=True).run() # receive only
# reliable stop-and-wait ARQ across two radios (RX first):
run_pair(sink_arq(rx_args="serial=30CD3F7", scheme="QPSK", fec=True),
         source_arq(tx_args="serial=30CD424", scheme="QPSK", fec=True))
print(tx(scheme="QPSK").command()) # just print the command (don't run)
```

The roles

Role	Meaning
tx / rx	transmit only / receive only (one radio each)
both	transmit and receive in one process (single-box loopback)
source_arq / sink_arq	the two ends of reliable stop-and-wait ARQ (with ACK)
sense	listen and report channel power (used by <code>channel_sense.py</code>)

The options you'll actually use (curated — the full list is `PARAMETERS.md`, or `sdr_system --help`)

Option	Example	Meaning
<code>--role</code>	<code>source_arq</code>	what this process does (table above)
<code>--scheme</code>	<code>QPSK</code>	modulation: BPSK QPSK 8-PSK 16/64-QAM, differential DBPSK
<code>--waveform</code>	<code>sc</code>	<code>sc</code> (single-carrier) or <code>ofdm</code> (many subcarriers, tracks frequency)
<code>--fec</code>	<code>true</code>	turn error-correction on
<code>--fec-type</code>	<code>turbo</code>	<code>conv</code> (fast), <code>ldpc</code> , or <code>turbo</code> (strongest)
<code>--fec_soft</code>	<code>true</code>	soft-decision decoding (better; pair with <code>ldpc/turbo</code>)
<code>--tx_args / --rx_args</code>	<code>serial=30CD424 / addr=192.168.20.2</code>	pick the radio (B210 serial / N210 addr)
<code>--tx_freq / --rx_freq</code>	<code>915e6</code>	carrier frequency (Hz)
<code>--tx_rate / --rx_rate</code>	<code>2e6</code>	sample rate (Hz); N210 needs <code>2e6</code>
<code>--symbol_rate</code>	<code>1e6</code>	symbol rate; N210 pairs <code>2e6/1e6</code>
<code>--tx_gain / --rx_gain</code>	<code>78 / 30</code>	amplifier gain (dB)
<code>--tx_subdev / --rx_subdev</code>	<code>A:A / A:0</code>	radio front-end (B210 A:A, N210 A:0)
<code>--ack_transport</code>	<code>tcp</code>	send the ARQ ACK over <code>tcp</code> (so an RX-only radio never transmits)
<code>--ack_host / --ack_port</code>	<code>192.168.1.50 / 5599</code>	where the ACK goes
<code>--bytes_length</code>	<code>125</code>	payload size per frame (bytes)
<code>--max_attempts</code>	<code>1</code>	retransmit limit; <code>1</code> = single-shot, <code>0/high</code> = keep trying
<code>--viz / --viz_dir</code>	<code>true / phy_outputs</code>	capture TX/RX signals + auto-plot to <code>phy_outputs/<scheme></code>
<code>--config <file></code>	<code>phy.cfg</code>	load all options from a file (CLI still overrides)

Two rules that save hours: **start the receiver before the transmitter**, and on free-running radios use **DQPSK** (single-carrier) or **OFDM** — plain coherent QPSK delivers poorly without a shared clock.

From a JSON file (`drivers/usrp/python/run.py`)

```
python3 drivers/usrp/python/run.py --list           # ready-made configs
python3 drivers/usrp/python/run.py configs/qpsk_arq_pair.json --dry-run # print, don't run
```

6. The building blocks (pyphy)

Every DSP stage is also a plain Python function, so you can compose the chain yourself (GNU-Radio style). Build it with `drivers/usrp/bindings/build.sh`; import with `PYTHONPATH=drivers/usrp/bindings python3` (add `arch -x86_64` on macOS).

Group	Functions
Framing	<code>frame</code> , <code>unframe</code>
FEC	<code>fec_encode</code> , <code>fec_decode</code> , <code>fec_decode_soft</code> , <code>fec_encoded_len</code> (<code>conv ldpc turbo</code> , <code>k</code>)
Modulation	<code>modulate</code> , <code>demodulate</code> , <code>soft_llr</code>
Single-carrier	<code>rrc_tx</code> , <code>rrc_rx</code> (<code>pulse shape / matched filter</code>)
Sync	<code>preamble</code> , <code>acq</code> , <code>cfo_correct</code> , <code>phase_correct</code>
OFDM	<code>ofdm_mod</code> , <code>ofdm_demod</code> , <code>ofdm_data_per_sym</code>
Radio (UHD build)	<code>Radio(role, args, freq, rate, symbol_rate, gain, subdev, ant) .transmit() .capture()</code>

```
import numpy as np, pyphy
bits = (np.random.rand(2000) > 0.5).astype(np.uint8)
syms = pyphy.modulate(bits, "QPSK")           # bits -> symbols
syms = syms * 0.6                             # <- your own op between stages
wave = pyphy.rrc_tx(syms, sps=2, beta=0.25)    # pulse-shape to a waveform
# ... channel/radio ...
back = pyphy.demodulate(pyphy.rrc_rx(wave, sps=2, beta=0.25)[:2], "QPSK")
```

Worked flowgraph: `drivers/usrp/python/phy_flow_example.py`.

7. Function cheat-sheet (the Python API)

Uniform API — `union/phy_link.py`

- `adapt(obj, role)` — wrap ANY object with `transmit()/receive()` into the internal app.
- `PayloadSpec(dtype, shape)` — declare an output's type + shape.
- `Codec.pack(arr) / Codec.unpack(buf)` — numpy array <-> self-describing bytes.
- `IdealChannel / PyphyChannel(scheme, fec, snr_db)` — radio-free channels.
- `run_loopback(tx, rx, channel, steps)` — radio-free round-trip driver.
- `RadioRoundTrip(role, ...).step(app)` — the two-host radio round-trip.

Radio wrapper — `drivers/usrp/python/sdr.py`

- `SDR(**opts)` — one invocation; `.command()` (print), `.run()` (blocking), `.popen()` (background), `.set(**opts)`.
- helpers `tx / rx / both / source_arq / sink_arq / run_pair / options`.

Spectrum — `drivers/usrp/python/channel_sense.py`, `freq_scan.py`

- `sense_channel(...)` — measure power at one frequency.
- `python3 drivers/usrp/python/freq_scan.py --start 902 --stop 928 --rx-args addr=192.168.20.2` — find a quiet carrier.

8. Troubleshooting (beginner gotchas)

Symptom	Fix
pyphy import / Python.h / arch error (macOS)	run via ./run.sh (it sets PYTHONPATH + arch -x86_64), or build with drivers/us
Exec format error on sdr_system	rebuild on that machine: cd phy && cmake -S . -B build && cmake --build b
receiver gets nothing / no ACK	start the receiver first ; keep both radios on; use DQPSK or OFDM
startup "rate check" fails	N210 needs --rx-rate 2e6 --symbol-rate 1e6; set --tx-rate = --rx-rate
a stuck receiver won't stop	press Ctrl-C again; last resort Ctrl-\ or kill -9
coherent QPSK ~17% delivered	free-running clocks — use DQPSK (single-carrier) or OFDM

Beyond this guide

For deeper reference (in the repo root / experiments/): **PARAMETERS.md** (every option — or run sdr_system --help), **SYSTEM_REFERENCE.pdf** (the engine math + every algorithm), **COMMANDS.md** and **USRP_CARRIER_MODULATION.txt** (ready-to-run command recipes per scheme / per device), **HARDWARE.md** (hardware setup), and **experiments/EXPERIMENT_GUIDE.pdf** (running the example apps).